Expt. No. 7

Title: To design the layout of the D latch using micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

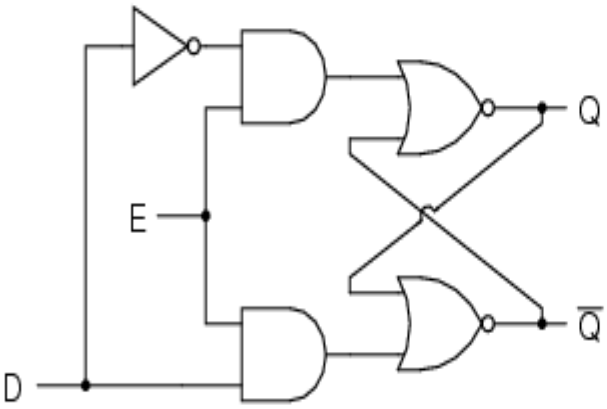
Experiment No. 7

Aim: To design the layout of the D latch using micro-wind.

Software Required: DSCH, Micro-wind software

Hardware: computer

Circuit Diagram:



E	D	Q	$\overline{Q}$
0	0	latch	latch
0	1	latch	latch
1	0	0	1
1	1	1	0

Theory:

Latch is an electronic device that can be used to store one bit of information. The D latch is used to capture, or 'latch' the logic level which is present on the Data line when the clock input is high. If the data on the D line changes state while the clock pulse is high, then the output, Q, follows the input, D. When the CLK input falls to logic 0, the last state of the D input is trapped and held in the latch.

CMOS circuit can be implemented using either logic diagram or
Structured description and direct transition shown A physical layer this can be created by 2 NOR cell adding a custom layout could be probably assume all area enable control can be added to basic

1. N-well/p-well spacing = 6A
2. N-well p-well width = 6A
3. Active spacing = 3A
4. Metal width = 2A
5. Metal to metal = 3A
6. Poly width = 3A
7. Mask = 3A

Procedure:

1] Start micro-wind software for layout.

2] Draw the D Latch following the design.

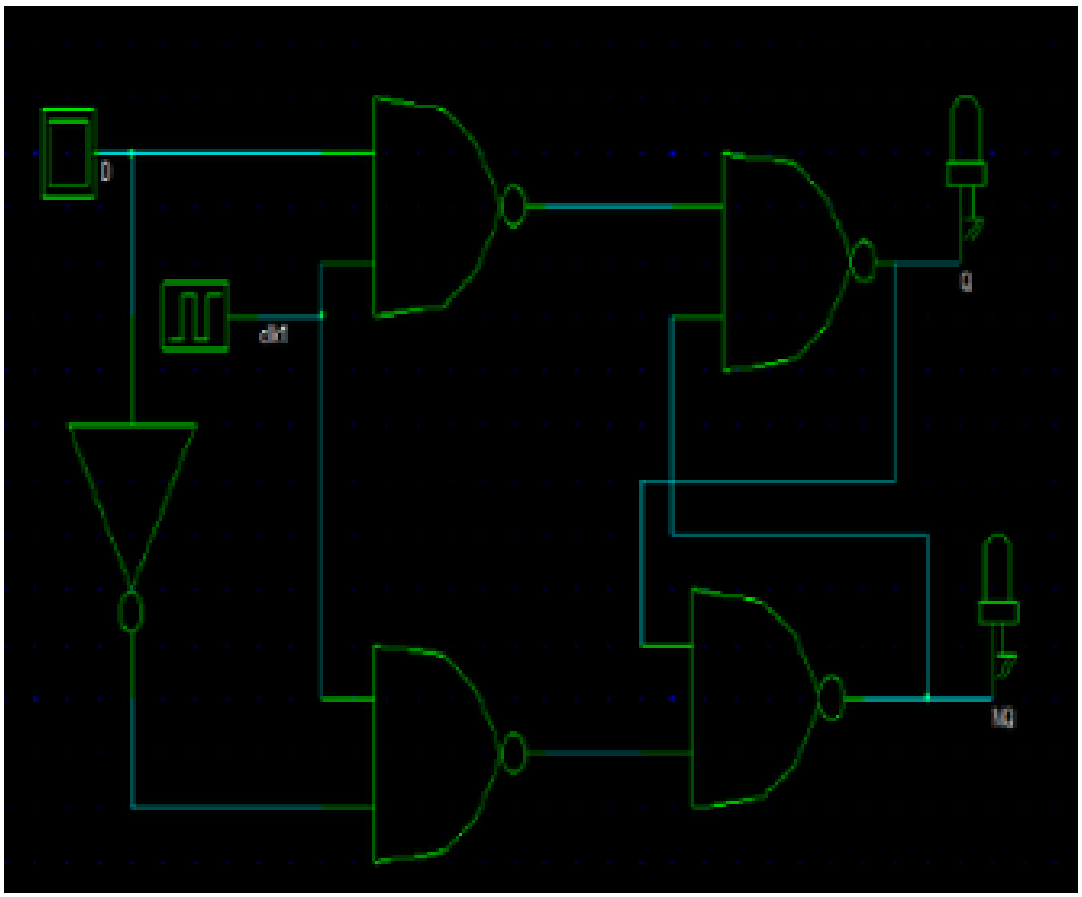
3] Click on DRC at every stage of design to avoid error.

4] Observe various output while using simulation output.

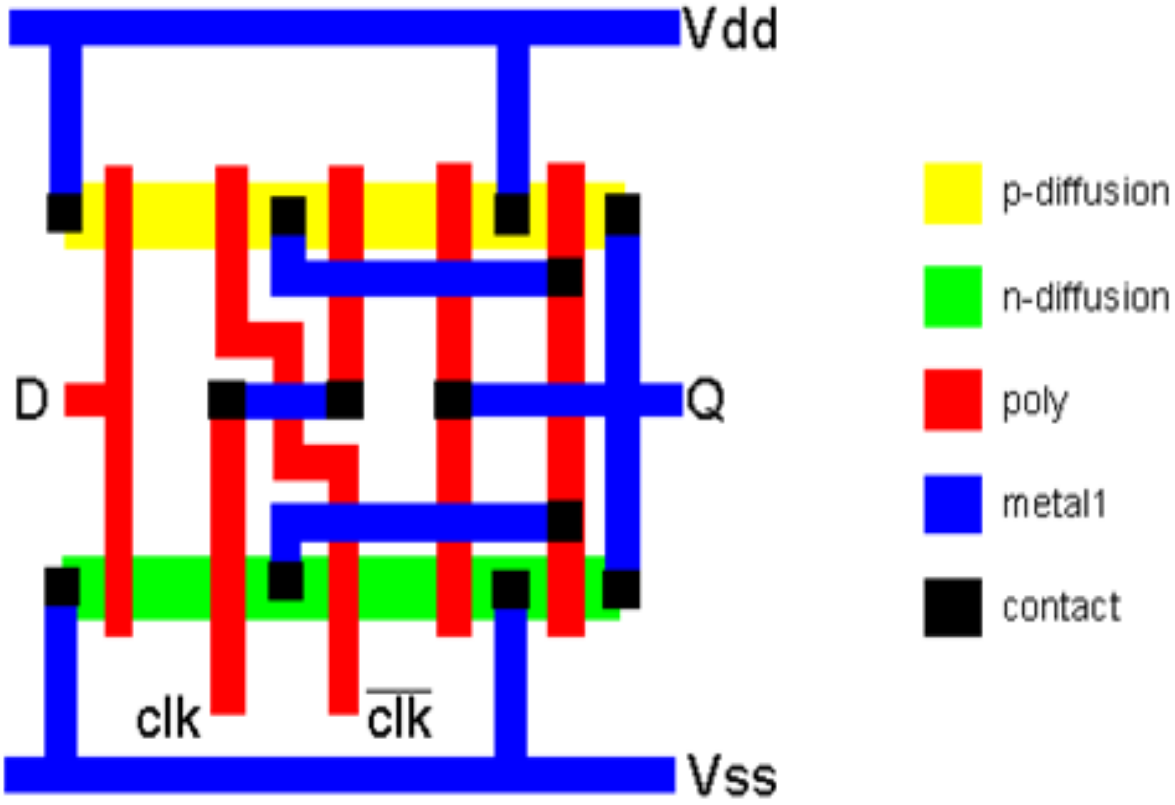
5] Software design and finally run a DRC.

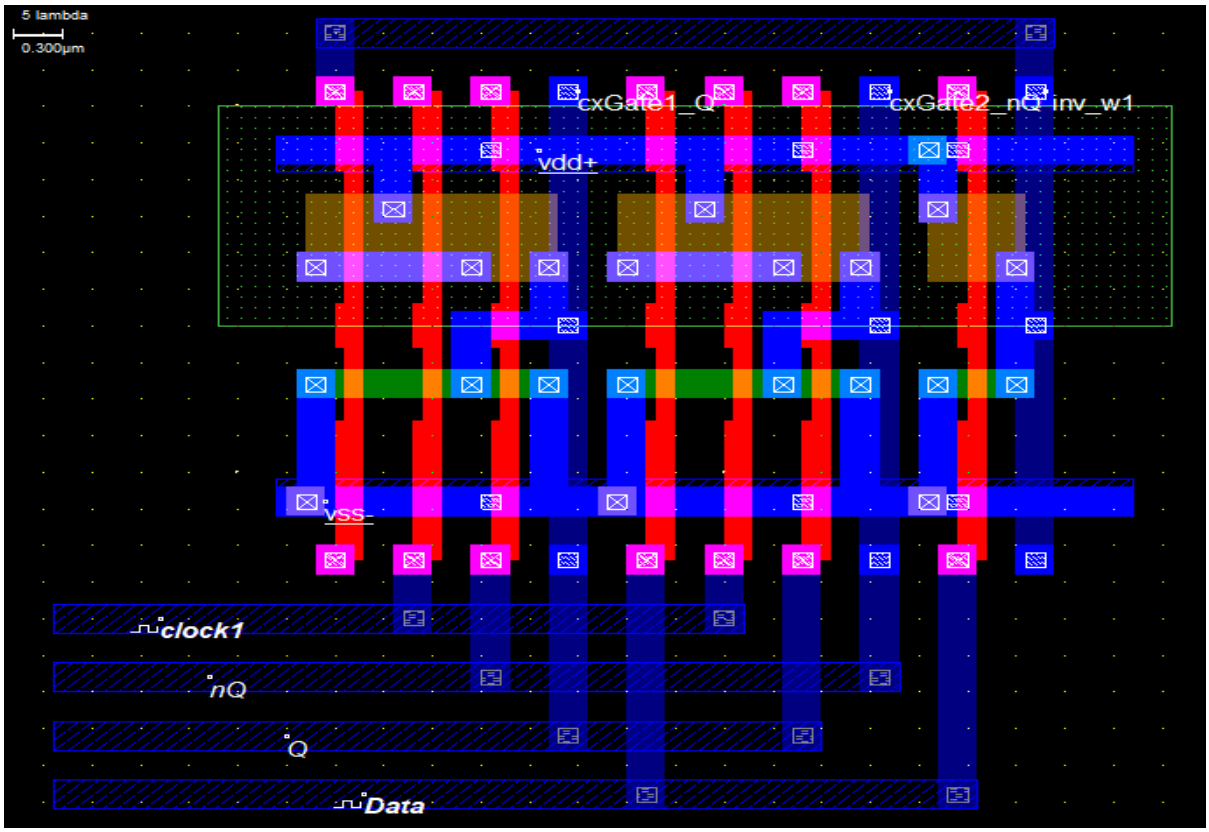
6] Record the output waveform observed.

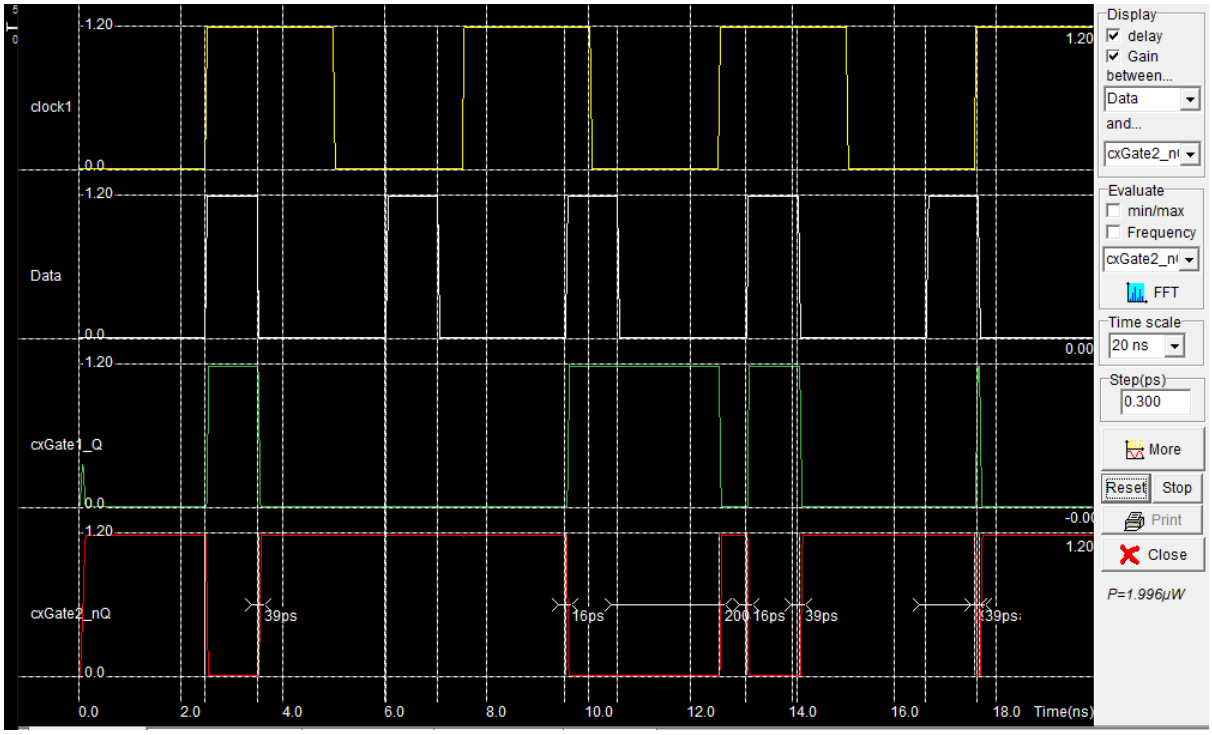
DSCH:

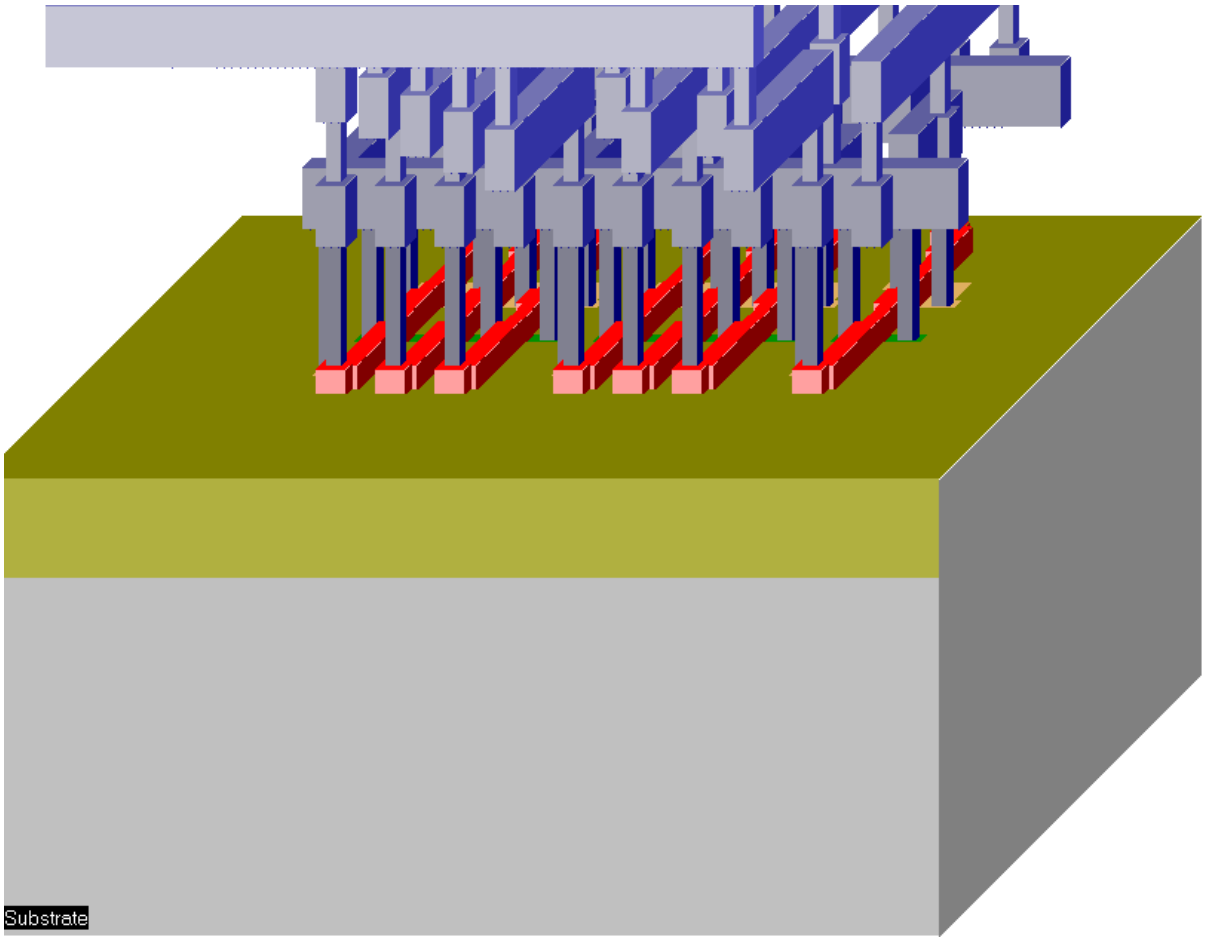


Layout:


Vdd
D
Q
clk
clk
Vss
p-diffusion
n-diffusion
poly
metal1
contact


5 lambda
0.300µm
cxGate1_Q
cxGate2_nQ
inv_w1
vdd
vss
clock1
nQ
Q
Data


clock1
Data
cxGate1_Q
cxGate2_nQ
1.20
0.0
39ps
16ps
200
16ps
39ps
39ps
0.0
2.0
4.0
6.0
8.0
10.0
12.0
14.0
16.0
18.0
Time(ns)
Display
delay
Gain
between...
Data
and...
cxGate2_nQ
Evaluate
min/max
Frequency
FFT
Time scale
20 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=1.996µW


Substrate

Pre Lab Questions:

1. Which flip-flop plays a vital role by functioning as the basic building block of a ripple counter?
2. Is D Flip flop is level Triggered?

Post Lab Questions:

3. How to construct D Flip flop?

Conclusion: Thus, we have designed layout of D Latch using Micro wind and verified layout using design rules.

CO's covered:

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